

COSC 594

Policy Optimization (IV)

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Outline

- AI SAFETY
- CONSTRAINED POLICY OPTIMIZATION
- NATURAL POLICY GRADIENT PRIMAL-DUAL METHODS

Outline

■ AI SAFETY

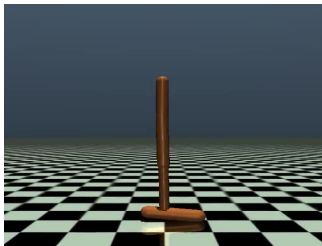
AI Safety

■ SAFETY CONSIDERATIONS

- ★ behave as humans intend
- ★ operate with uncertainty

■ SAFE RL

MuJoCo Hopper



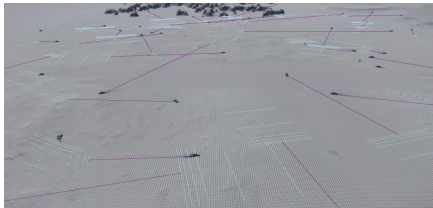
Goal forward moving

Requirements obstacle avoidance
energy budget
smoothness
⋮

Required behavior is more than a reward

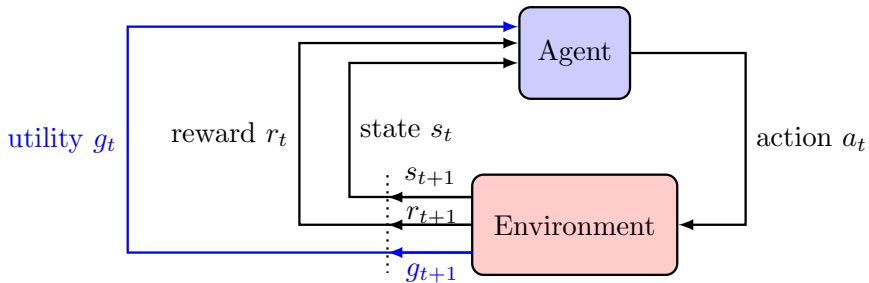
$$\begin{aligned} r(s, a) = & w_0 b_1(s, a) && \text{path following} \\ & + w_1 b_2(s, a) && \text{speed limit} \\ & + w_2 b_3(s, a) && \text{energy budget} \\ & + w_3 b_4(s, a) && \text{acceleration limit} \\ & \vdots && \simeq 400 \text{ lines of code, but hopefully work!} \end{aligned}$$

Unforeseen behavior in navigation



RL framework

■ CONSTRAINED MDP



Autonomous driving (Waymo)



Goal reaching a destination

Requirements collision avoidance
energy budget

⋮

Outline

- AI SAFETY
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Constrained policy optimization

Policy: $\pi : S \text{ (states)} \rightarrow \Delta_A \text{ (action distribution)}$

Reward value function: $V_r^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$

Utility value function: $V_g^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t g(s_t, a_t) \mid s_0 \sim \rho \right]$

■ NONCONVEX POLICY OPTIMIZATION

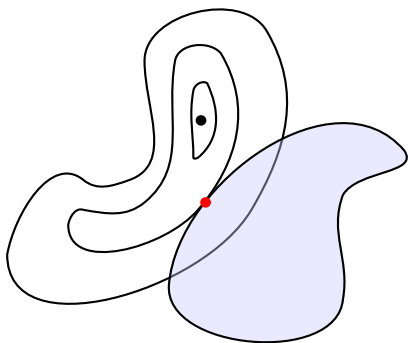
maximize $V_r^\pi(\rho)$ $\longrightarrow$ **standard goal**

subject to $V_g^\pi(\rho) \geq 0$ $\longrightarrow$ **policy constraint**

★ Limit the policy space to **an inequality constraint**

■ 2-D LEVEL NONCONVEX CURVES

optimal, but in**feasible**



feasible region

Optimality w/ **feasibility**

Lagrangian approach

Lagrangian: $L(\pi, \lambda) = V_r^\pi(\rho) + \lambda V_g^\pi(\rho)$

Lagrange multiplier: $\lambda (\geq 0)$

Composite reward: $r + \lambda g$

■ NONCONVEX POLICY OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V_{r + \lambda g}^\pi(\rho)$$

Multiple solutions, often infeasible

Strong duality

Dual problem: $\underset{\lambda}{\text{minimize}} \underset{\pi}{\text{maximize}} \quad V_{r+\lambda g}^{\pi}(\rho)$

Primal problem: $\underset{\pi}{\text{maximize}} \underset{\lambda}{\text{minimize}} \quad V_{r+\lambda g}^{\pi}(\rho)$

■ LINEAR PROGRAMMING

$$\begin{array}{ll} \underset{d_{\rho}^{\pi}(\cdot, \cdot)}{\text{maximize}} & \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim d_{\rho}^{\pi}(\cdot, \cdot)} [r(s, a)] \\ \text{subject to} & \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim d_{\rho}^{\pi}(\cdot, \cdot)} [g(s, a)] \geq 0 \end{array}$$

★ Zero duality gap

Primal-dual method

Dual problem: $\underset{\lambda}{\text{minimize}} \underset{\pi}{\text{maximize}} V_{r+\lambda g}^{\pi}(\rho)$

Primal problem: $\underset{\pi}{\text{maximize}} \underset{\lambda}{\text{minimize}} V_{r+\lambda g}^{\pi}(\rho)$

■ NONCONVEX MIN-MAX POLICY OPTIMIZATION

Strong duality $\implies$ **exchange** of min and max

$$\underset{\lambda}{\text{minimize}} \underset{\pi}{\text{maximize}} V_{r+\lambda g}^{\pi}(\rho) = \underset{\pi}{\text{maximize}} \underset{\lambda}{\text{minimize}} V_{r+\lambda g}^{\pi}(\rho)$$

OBJECTIVE: find a minimax point (π^*, λ^*)

■ PRIMAL-DUAL UPDATE

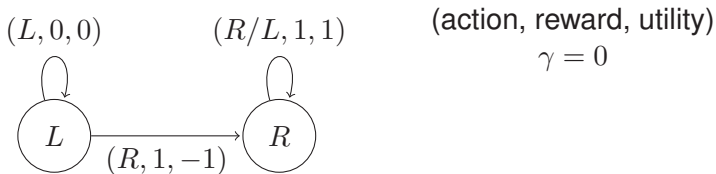
Policy ascent direction: $\nabla_{\pi} V_{r+\lambda g}^{\pi}(\rho)$

Dual descent direction: $V_g^{\pi}(\rho)$

$$\pi^+ \leftarrow \pi + \eta_1 \nabla_{\pi} V_{r+\lambda g}^{\pi}(\rho)$$

$$\lambda^+ \leftarrow \lambda - \eta_2 V_g^{\pi}(\rho)$$

Oscillation example



$(\rho, 1 - \rho)$ – initial state distribution

$(p, 1 - p)$ in state L – policy

$$V_r^p(\rho) = (1 - p)\rho + (1 - \rho)$$

$$V_g^p(\rho) = -(1 - p)\rho + (1 - \rho)$$

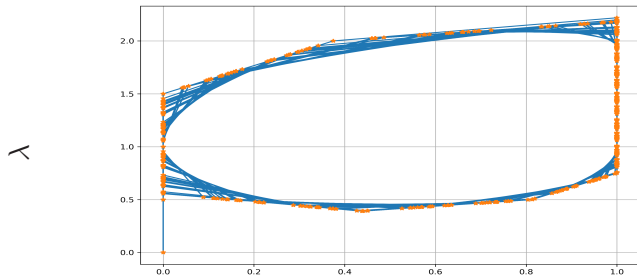
★ Optimal policy

$$p \geq \frac{2\rho - 1}{\rho} \text{ for } \rho \in \left[\frac{1}{2}, 1\right] \implies p^* = \frac{2}{3} \text{ when } \rho = \frac{3}{4}$$

Oscillation example

$$L(p, \lambda) = \underbrace{(1-p) \frac{3}{4} + \frac{1}{4}}_{V_r^P(\rho)} + \lambda \underbrace{\left(-(1-p) \frac{3}{4} + \frac{1}{4} \right)}_{V_g^P(\rho)}$$

bilinear Lagrangian



$$p \text{ (} p^* = 0.66 \text{)}$$

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Primal-dual version of natural policy gradient

Policy ascent direction: $F_{\rho}(\theta)^{\dagger} \nabla_{\theta} V_{r+\lambda g}^{\pi_{\theta}}(\rho)$

Dual descent direction: $V_g^{\pi}(\rho)$

■ PRIMAL-DUAL UPDATE

$$\begin{aligned}\theta^+ &= \theta + \eta_1 F_{\rho}(\theta)^{\dagger} \nabla_{\theta} V_{r+\lambda g}^{\pi_{\theta}}(\rho) \\ \lambda^+ &= \operatorname{argmin}_{\lambda' \geq 0} \lambda' V_g^{\pi}(\rho) + \frac{1}{2\eta_2} (\lambda' - \lambda)^2\end{aligned}$$

Costly inverse of Fisher information matrix

Parallel b/w natural gradient and action value function

Softmax policy: $\pi_{\theta}(a | s) = \frac{e^{\theta_{s,a}}}{\sum_{a'} e^{\theta_{s,a'}}}$ w/ parameter $\theta \in \mathbb{R}^{|S||A|}$

Linear regression: $w^{\star} = \text{LR} \left(w; \nabla_{\theta} \log \pi_{\theta}, Q_{r+\lambda g}^{\pi_{\theta}} \right)$

■ MULTIPLE OPTIMAL SOLUTIONS

$$\begin{aligned} w^{\star} &= F_{\rho}(\theta)^{\dagger} \nabla_{\theta} V_{r+\lambda g}^{\pi_{\theta}}(\rho) \\ &\simeq Q_{r+\lambda g}^{\pi_{\theta}} \end{aligned}$$

$F_{\rho}(\theta)^{\dagger} \nabla_{\theta} V_{r+\lambda g}^{\pi_{\theta}}(\rho)$ is parallel with $Q_{r+\lambda g}^{\pi_{\theta}}$

Natural policy gradient primal-dual method

Policy ascent direction: $Q_{r+\lambda g}^{\pi_\theta}$

Dual descent direction: $V_g^\pi(\rho)$

■ POLICY PRIMAL-DUAL UPDATE

$$\begin{aligned}\pi^+(\cdot | s) &= \operatorname{argmax}_{\pi'} \langle \pi'(\cdot | s), Q_{r+\lambda g}^\pi(s, \cdot) \rangle - \frac{1}{\eta_1} D_{\text{KL}}(\pi'(\cdot | s) \| \pi(\cdot | s)) \\ \lambda^+ &= \operatorname{argmin}_{\lambda' \geq 0} \lambda' V_g^\pi(\rho) + \frac{1}{2\eta_2} (\lambda' - \lambda)^2\end{aligned}$$

★ Explicit policy update $\pi^+(\cdot | s) \propto \pi(\cdot | s) e^{\eta_1 Q_{r+\lambda g}^\pi(s, \cdot)}$

Convex policy update

Global convergence

■ OPTIMALITY GAP

$$\frac{1}{T} \sum_{t=0}^{T-1} (V_r^*(\rho) - V_r^{(t)}(\rho)) = O\left(\frac{1}{(1-\gamma)^2} \frac{1}{\sqrt{T}}\right)$$

■ CONSTRAINT VIOLATION

$$\left[\frac{1}{T} \sum_{t=0}^{T-1} (b - V_g^{(t)}(\rho)) \right]_+ = O\left(\frac{1}{(1-\gamma)^2} \frac{1}{\sqrt{T}}\right)$$

- ★ $O(\cdot)$ – dimension-free: no $|S|$, $|A|$, and ρ
- ★ $O(\frac{1}{\epsilon^2})$ – iteration complexity for ϵ -near optimal policy

Convergence analysis

Step #1: performance difference & telescope KL

$$V_r^\star(\rho) - V_r^{(t)}(\rho)$$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^\star} \left[\sum_{a \in A} \pi^\star(a | s) A_r^{(t)}(s, a) \right]$$

$$\leq \frac{1}{\eta_1} \mathbb{E}_{s \sim d^\star} \left[D_{\text{KL}}(\pi^\star(\cdot | s), \pi^{(t)}(\cdot | s)) - D_{\text{KL}}(\pi^\star(\cdot | s), \pi^{(t+1)}(\cdot | s)) \right]$$

$$- \lambda^{(t)}(V_g^\star(\rho) - V_g^{(t)}(\rho))$$

$$+ \frac{1}{\eta_1} \mathbb{E}_{s \sim d^\star} \log Z^{(t)}(s)$$

$$\frac{1}{\eta_1 T} \sum_{t=0}^{T-1} \mathbb{E}_{s \sim d^\star} \log Z^{(t)}(s) \lesssim \frac{1}{\sqrt{T}}$$

■ AVERAGE PERFORMANCE

$$V_r^*(\rho) - \frac{1}{T} \sum_{t=0}^{T-1} V_r^{(t)}(\rho) + \lambda \left(V_g^*(\rho) - \frac{1}{T} \sum_{t=0}^{T-1} V_g^{(t)}(\rho) \right) \lesssim \frac{1}{\sqrt{T}}$$

any $\lambda \in [0, C]$, $C > 0$

$$V_g^*(\rho) \geq b$$

Step #2: linear programming & strong duality

■ CONSTRAINED OPTIMALITY MEASURE

$$\exists \pi', \underbrace{V_r^*(\rho) - V_r^{\pi'}(\rho)}_{\text{optimality gap}} + C \times \underbrace{[b - V_g^{\pi'}(\rho)]_+}_{\text{constraint violation}} \lesssim \frac{1}{\sqrt{T}}$$

Summary

■ AI SAFETY

Safe RL via constraints

■ CONSTRAINED POLICY OPTIMIZATION

Nonconvex optimization problem

■ NATURAL POLICY GRADIENT PRIMAL-DUAL METHODS

Primal-dual version of natural policy gradient methods, global best-iterate convergence